Syed Naved Mehdi

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Education

SRM Institute of Science and Technology — Chennai, India
M.Tech – Computer Science & Engineering (AI & Machine Learning) — *2025 – 2027*
- Current SGPA: **8.60 / 10.0**

Sarala Birla University — Ranchi, Jharkhand
B.Tech – Artificial Intelligence — *2021 – 2025*
- CGPA: 7.1 / 10.0

Technical Skills

Languages: Python, Java, SQL
ML / DL Frameworks: PyTorch, TensorFlow, Scikit-learn, Keras
Generative AI: Large Language Models (LLMs), RAG Architecture, Agentic AI, FAISS/ChromaDB, Prompt Engineering
Data & Computer Vision: NumPy, Pandas, OpenCV, MediaPipe, ResNet, Transformers
Deployment & Backend: FastAPI, Flask, REST APIs, Docker, Git, GitHub Actions (CI/CD)

Engineering Projects

Production-Grade Legal RAG Microservice | *Python, FastAPI, RAG, ChromaDB, Docker* — Patent App: 202541106145
- Developed an AI legal assistant utilizing Retrieval-Augmented Generation (RAG) to ground LLM responses against verified motor vehicle law documents, significantly reducing model hallucinations.
- Implemented dense vector retrieval and semantic chunking pipelines to extract relevant legal context before injecting it into the LLM context window.
- Containerized the microservice using Docker for consistent local deployment and scalable environment isolation; patent application formally published by the Indian Patent Office.

Hardware-Level Gesture Controller | *Python, MediaPipe, OS-level Ctypes* — GitHub
- Engineered a multi-modal Human-Computer Interaction (HCI) pipeline mapping MediaPipe hand-tracking landmarks directly to OS-level DirectX hardware inputs via `ctypes` for zero-latency execution.
- Designed finite state-machine logic and hysteresis buffering to eliminate input jitter and prevent gesture spamming during continuous execution.
- Implemented custom software Pulse Width Modulation (PWM) emulation to translate binary keyboard strokes into proportional analog driving controls based on spatial finger tracking.

Real-Time Gaze Heatmap & Attention Tracker | *Python, OpenCV, MediaPipe, NumPy* — GitHub
- Built a cross-platform computer vision pipeline tracking iris coordinates and nose tips to dynamically generate spatial accumulating Gaussian heatmaps.
- Developed a screen calibration module utilizing `cv2.findHomography` to map webcam coordinates to physical screen dimensions with sub-pixel accuracy.
- Optimized spatial stability by applying Exponential Moving Average (EMA) smoothing and head-movement suppression algorithms to the raw gaze vector outputs.

Research & Publications

Attn. Multi-Resolution Deep Learning for Electricity Load Forecasting | *ICICDA 2026*
- Published at ICICDA 2026 (SRM IST Vadapalani) – proposed MR-Attn-LSTM, a novel parallel-branch LSTM with Bahdanau attention processing hourly, daily, and weekly temporal resolutions.
- Trained on 43,776 hours of real TNEB & KSEB operational data for simultaneous 1h/6h/24h multi-horizon forecasting, achieving MAPE 1.50% (Tamil Nadu) / 1.35% (Kerala) with a 26–27% MAE reduction over baselines.

Brain Tumour Classification using Transfer Learning | *ICMINT 2025*
- Published at ICMINT 2025 – Co-authored research evaluating an MRI-based tumour detection system using a fine-tuned ResNet Transformer with transfer learning (ImageNet pretraining) to address limited medical dataset constraints.

Awards & Certifications

- **Gold Medal (Rank #1 in Madhya Pradesh)** – National Science Olympiad (NSO)
- **Building Agentic AI Applications with LLMs** – NVIDIA
- **Brain Computer Interface using Deep Learning** – Udemy
- **Introduction to Data Analytics** – Internshala
- **Interpersonal Skills** – US English